

SCHEDULING ALGORITHMS :

1. Fcfs :

```
madhumitha@localhost:~$ ls
ABC Downloads madhu.sh pidoutput sample.txt
abc.sh emp.txt maith Public sjf
ab.h exec.c mn rahuk.sh sujje
algo fork.c Music rahul Templates
a.out kee opendir.c rahul.sh Videos
'banks alg0'$'\n''mkdir banks' mad os_lab REC vms
Desktop mad3 os_lab.c riya XYZ
dir.c madd Pictures riyaa.sh
Documents madhu pid.c sample.c

[madhumitha@localhost ~]$ vi os_lab1.c
[madhumitha@localhost ~]$ gcc os_lab1.c -o os_lab1
[madhumitha@localhost ~]$ ./os_lab1
Enter number of processes: 3
Enter Arrival Time and Burst Time for P1: 0 5
Enter Arrival Time and Burst Time for P2: 1 3
Enter Arrival Time and Burst Time for P3: 2 8

Average Turnaround Time: 8.67
Average Waiting Time: 3.33
[madhumitha@localhost ~]$
```



madhumitha@localhost:~ — /usr/bin/vim os_lab1.c

```
[madhumitha@localhost ~]$ ls
- int main() {
-     int n, r, j; j;
-
-     printf("Enter number of processes: ");
-     scanf("%d", &n);
-
-     int at[n], bt[n], ct[n], tat[n], wt[n];
-
-     // Input
-     for (int i = 0; i < n; i++) {
-
-         printf("Enter Arrival Time and Burst Time for P%d: ", i+1);
-         scanf("%d %d", &at[i], &bt[i]);
-     }
-
-     float total_tat = 0, total_wt = 0;
-
-     // First process
-     ct[0] = at[0] + bt[0];
-
-     // Remaining processes
-     for (int i = 1; i < n; i++) {
-         if(ct[i-1] < at[i])
-             ct[i] = at[i] + bt[i]; // CPU Idle
-         else
-             ct[i] = ct[i-1] + bt[i];
-     }
-
-     // Calculate TAT and WT
-     for (int i = 0; i < n; i++) {
-         tat[i] = ct[i] - at[i];
-         wt[i] = tat[i] - bt[i];
-     }
-
-     printf("\nAverage Turnaround Time: %.2f", total_tat / n);
-     printf("\nAverage Waiting Time: %.2f\n", total_wt / n);
- } return 0;
```

2. sjf :

```
madhumitha@localhost:~  
[madhumitha@localhost ~]$ ls  
ABC Downloads madhu.sh pid.c sample.c  
abc.sh emp.txt maith pidoutput sample.txt  
ab.h exec.c mn Public sjf  
algo fork.c Music rahuk.sh sujje  
a.out kee opendir.c rahul Templates  
'banks alg0'$'\n''mkdir banks' mad os_lab rahul.sh Videos  
Desktop mad3 os_lab1.c REC vms  
dir.c madd os_lab.c riya XYZ  
Documents madhu Pictures riyaa.sh  
[madhumitha@localhost ~]$ vi sjf.c  
[madhumitha@localhost ~]$ vi sjf.c  
[madhumitha@localhost ~]$ gcc sjf.c -o sjf  
[madhumitha@localhost ~]$ ./sjf  
Enter number of processes: 3  
Enter Arrival Time and Burst Time for P1: 0 7  
Enter Arrival Time and Burst Time for P2: 2 4  
Enter Arrival Time and Burst Time for P3: 4 1  
  
Average Turnaround Time: 7.00  
Average Waiting Time: 3.00  
[madhumitha@localhost ~]$  
  
- int main()  
- {  
-     int n;  
-     printf("Enter number of processes: ");  
-     scanf("%d", &n);  
-  
-     int at[n], bt[n], ct[n], tat[n], wt[n];  
-  
-     int done[n] = 0;  
-  
-     int completed = 0;  
-     int completionTime = 0;  
-     float avgtat = 0, avgwt = 0;  
-  
-     // SJF Scheduling (Non-Preemptive)  
-     while(completed < n)  
-     {  
-         int idx = -1;  
-         int minBT = 9999;  
-  
-         for(int i = 0; i < n; i++)  
-         {  
-             if(at[i] <= completionTime && done[i] == 0)  
-             {  
-                 if(bt[i] < minBT)  
-                 {  
-                     minBT = bt[i];  
-                     idx = i;  
-                 }  
-             }  
-         }  
-  
-         if(idx != -1)  
-         {  
-             completionTime += bt[idx];  
-             done[idx] = 1;  
-             completed++;  
-             avgtat += tat[idx];  
-             avgwt += wt[idx];  
-         }  
-     }  
-  
-     printf("Average Turnaround Time: %.2f\n", avgtat/n);  
-     printf("Average Waiting Time: %.2f\n", avgwt/n);  
- }  
[madhumitha@localhost ~]
```

Round robin

```
madhumitha@localhost:~$ ls
ABC Downloads madhu.sh pid.c sample.c
abc.sh emp.txt maith pidoutput sample.txt
ab.h exec.c mn Public sjf.c
algo fork.c Music rahuk.sh sujje
a.out kee opendir.c rahul Templates
'banks algo'$'\n'$'mkdir banks' mad os_lab rahul.sh Videos
Desktop mad3 os_lab1.c REC vms
dir.c madd os_lab.c riya XYZ
Documents madhu Pictures riya.sh

[madhumitha@localhost ~]$ vi roundrobin.c
[madhumitha@localhost ~]$ gcc roundrobin.c -o roundrobin
[madhumitha@localhost ~]$ ./roundrobin
bash: ./roundrobin: No such file or directory
[madhumitha@localhost ~]$ ./roundrobin
Enter number of processes: 3
Enter Time Quantum: 2
Enter AT and BT for P1: 0 5
Enter AT and BT for P2: 1 3
Enter AT and BT for P3: 2 2

Average Turnaround Time: 6.33
Average Waiting Time: 3.33
[madhumitha@localhost ~]$
```

Priority scheduling

```
madhumitha@localhost:~  
[madhumitha@localhost ~]$ ls  
- banker.c Documents mait Mude Pictures sh whilew.sh  
a bankers Downloads maitre os_lab1 Public sjf XYZ  
ABC bm h mash os_lab1.c REC sjf.c  
abc.sh close mai withi os_lab1.c sample.c tesatstxt  
abc.sh desktop maithu os_lab1 roundrobin.c sample.txt whilew.sh  
shf Documents maitre os_lab1.c XYZ  
[madhumitha@localhost ~]$ vi priority  
[madhumitha@localhost ~]$ gcc priority.c o priority  
[madhumitha@localhost ~]$ ./priority  
Enter number of processes: 3  
  
Enter AT, BT and Priority for P1: 0 5 2  
  
Enter AT, BT and Priority for P2: 1 3 1  
  
Enter AT, BT and Priority for P3: 2 8 3  
2 8 3  
  
Average Turnaround Time: 8.67  
Average Waiting Time: 3.33  
[madhumitha@localhost ~]$
```